Vention Software Reliability Challenge

Context

Vention offers a Software platform to Design, Build and Automate Robotic Machines: vention.com.
The Engineering team is composed of around ~60 developers working in several shared code bases across the software stack.
Each team can ship at different speeds based on their criticality and reliability required by their areas.
Some teams work on the Machine Motion Hardware Controller Stack and others on the Vention Cloud platform.

We have two deployment targets for the software that we build:

- **Cloud** — contains design, simulation, configuration and ordering platform software modules. It's a two week release cadence, we own the runtime, rollback is possible.
- **Edge** — contains firmware and software modules running on controllers driving real motion in customer factories. Customer-owned, often disconnected, often months behind. No rollback on a machine mid-shift. Failures stop lines or create hazards. We can patch the software but it comes at a high cost.

1. Design a Process to drive Software Quality/Reliability

How would you structure Software Quality/Reliability for **Vention** so that we are successful in shipping quality products on both targets?
How do you ensure that the Software Artefacts work hand in hand with the Hardware at each release cycle?

We want you to produce a document explaining the process and answering:

- How do you measure quality?
- How to track quality over time?
- How to ensure that findings are injected back into the Software Development Life Cycle.
- How do you detect issues early, triage the defects, and track resolutions to maintain a high quality bar?

Bonus:

- An MMAI controller in the field intermittently loses EtherCAT communication. How would your process, in section 2 above, have caught this before release, and how would you prevent it from happening in the future?

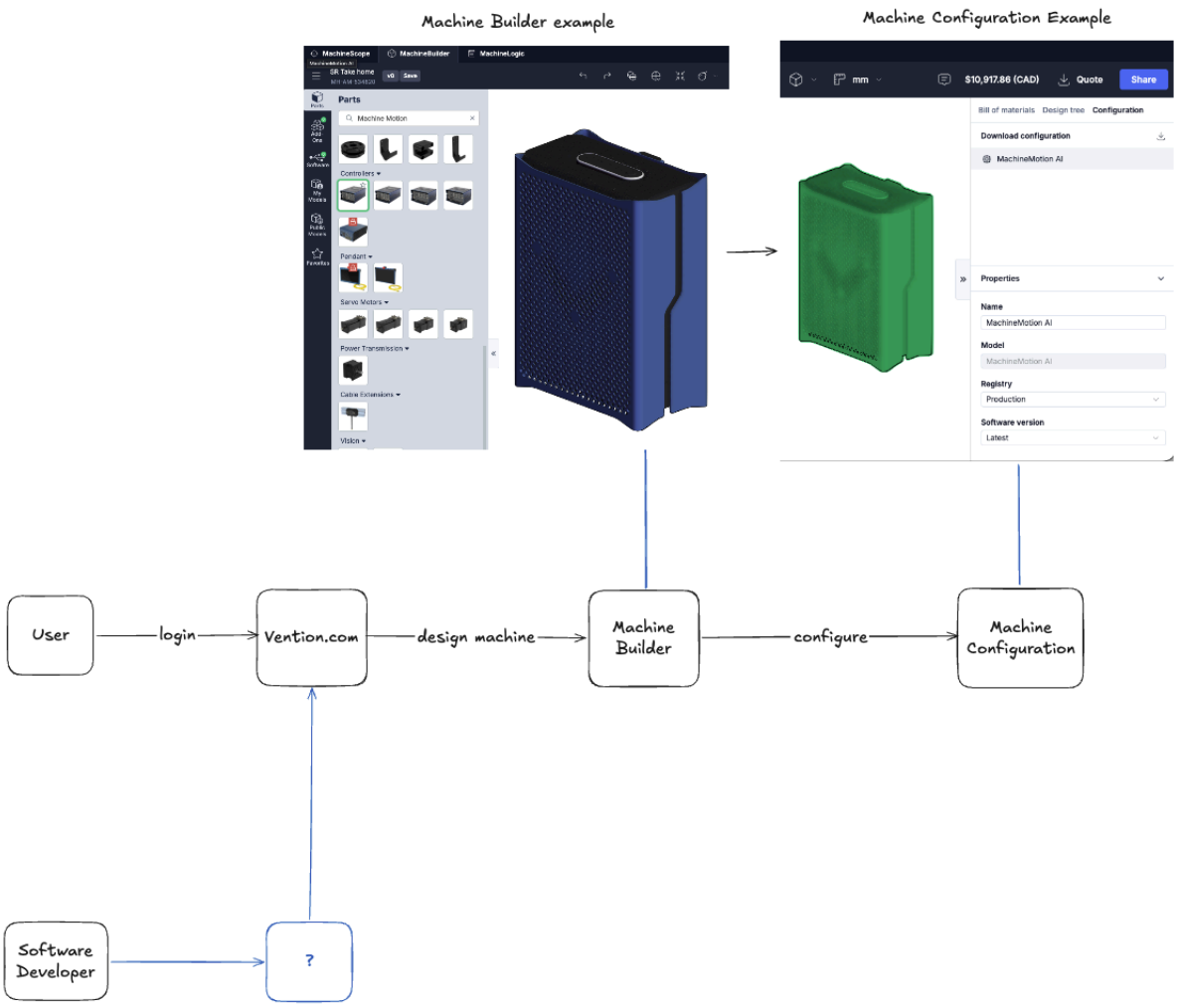
2. Cloud End-To-End Code Challenge

Objective

- Build a system that validates a vention.com feature through end-to-end tests, using a black-box approach.
- The system should validate that two Machine Motion Controllers cannot be added to the same Machine Builder design **to be automated**.

Scope

- The system must interact with vention.com's **Machine Builder**. As a machine is assembled, a Software Defined Configuration is generated automatically.
- You are targeting the property inspector for the Machine Motion AI Controller's Configuration.
- To understand what this feature does,
    1. Go to vention.com and create an account.
    2. Create a new design and add a Machine Motion Controller.
    3. Select the Machine Motion AI Controller (MMAI).
    4. Navigate to the Configuration Pane.
    5. You will see a property inspector for the controller in the right pane.
    6. Select the MMAI in the TreeView and the property pane will appear.



Deliverables

- A sample test repository named **platform-qa**, shared with us via a GitHub link.
- A CI pipeline designed to run against that repository.
- An end-to-end test system that treats vention.com strictly as a black box (see image below).

Guidelines

- Use any tools you like, including AI. We're looking for someone who can design this *and* explain it clearly in the next stage of the interview. If AI helped, tell us how.
- If you hit a roadblock, tell us what it was and how you would mitigate it.
- Do not use any confidential material from a current or former employer.